

Language Agent Tree Search

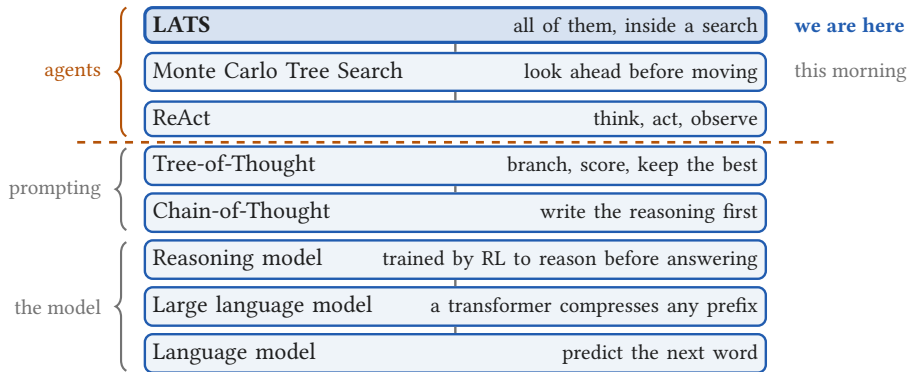


One frozen model as a planning agent

Weekend 6 · Frontier Agentic AI

Carlos Cotrini

Where LATS sits



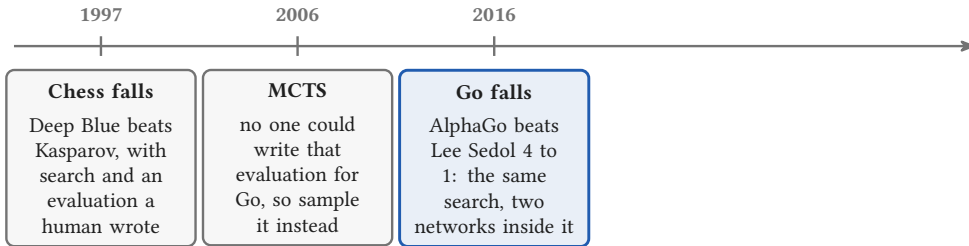
Monte Carlo Tree Search, before language models



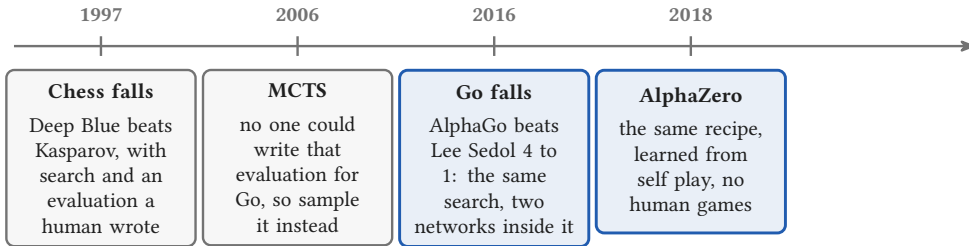
Monte Carlo Tree Search, before language models



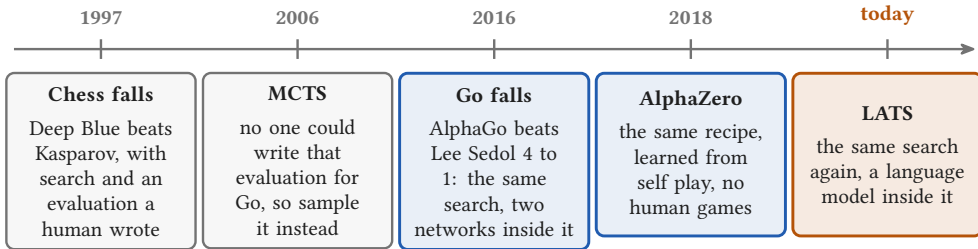
Monte Carlo Tree Search, before language models



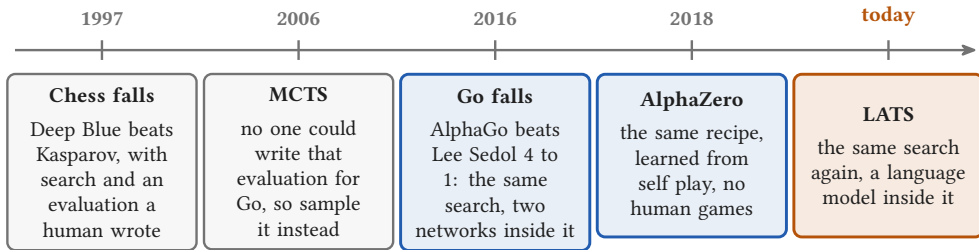
Monte Carlo Tree Search, before language models



Monte Carlo Tree Search, before language models



Monte Carlo Tree Search, before language models



The search itself barely changed across twenty years. What changed each time is what supplies the two things it needs: something to propose moves, and something to judge a position. LATS supplies both from one frozen model.

Coulom 2006; Kocsis and Szepesvari 2006; Silver et al., Nature 2016; Silver et al., Science 2018.

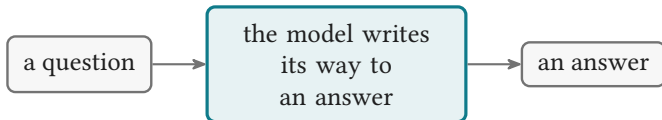


Section 1

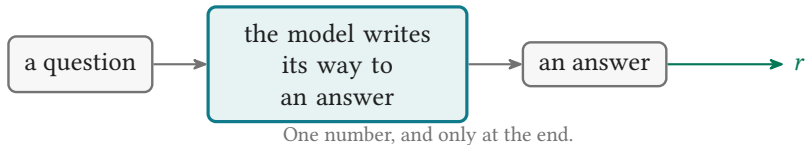
Recap: the fundamentals

1

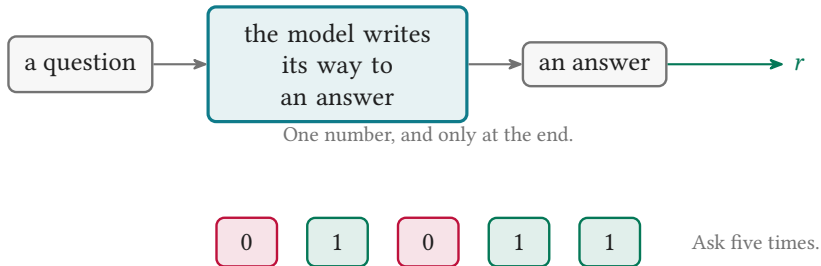
Reinforcement learning, for reasoning



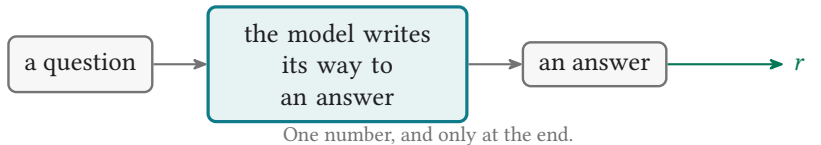
Reinforcement learning, for reasoning



Reinforcement learning, for reasoning

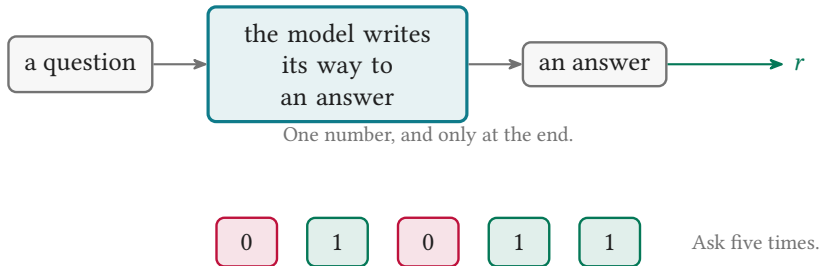


Reinforcement learning, for reasoning



The group's own average is the bar to clear.

Reinforcement learning, for reasoning

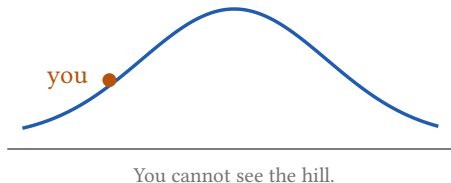


The group's own average is the bar to clear.

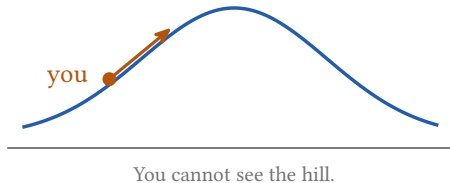
AIME 2024: **15.6** → **77.9** with no supervised fine tuning

DeepSeek-R1-Zero. Guo et al., Nature 645, 633 to 638, 2025.

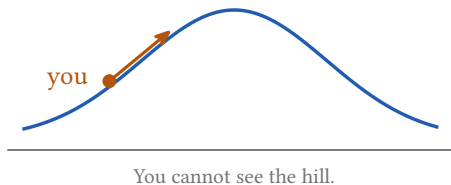
Gradient ascent, and the grasshopper



Gradient ascent, and the grasshopper

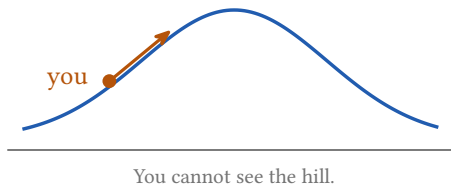


Gradient ascent, and the grasshopper



Go where the slope points.

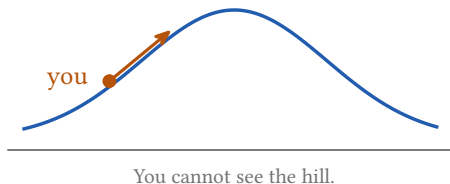
Gradient ascent, and the grasshopper



Go where the slope points.

Go further when it is steeper.

Gradient ascent, and the grasshopper

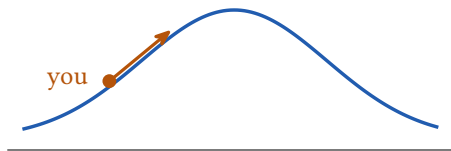


Go where the slope points.

Go further when it is steeper.

$$\theta \leftarrow \theta + \alpha \nabla j(\theta)$$

Gradient ascent, and the grasshopper



You cannot see the hill.

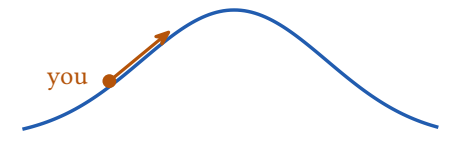
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On rough ground the slope is estimated, never measured. That is the case RL is in.

Gradient ascent, and the grasshopper



You cannot see the hill.

Go where the slope points.

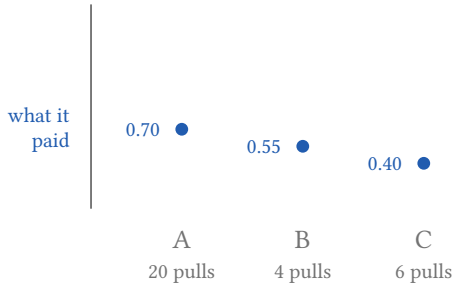
Go further when it is steeper.

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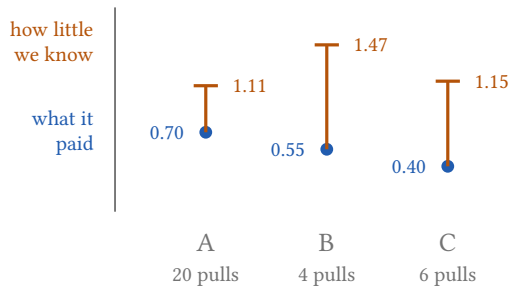
On rough ground the slope is estimated, never measured. That is the case RL is in.

Play it

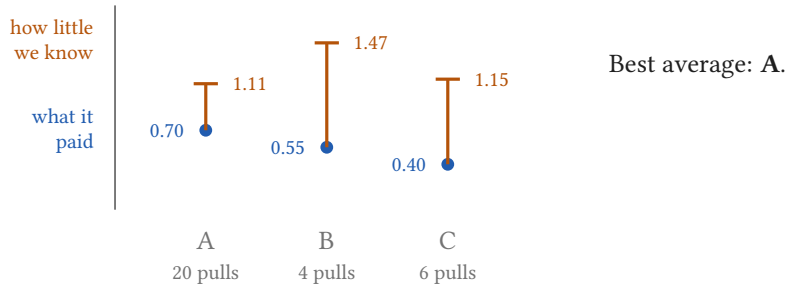
Bandits, and the upper bound



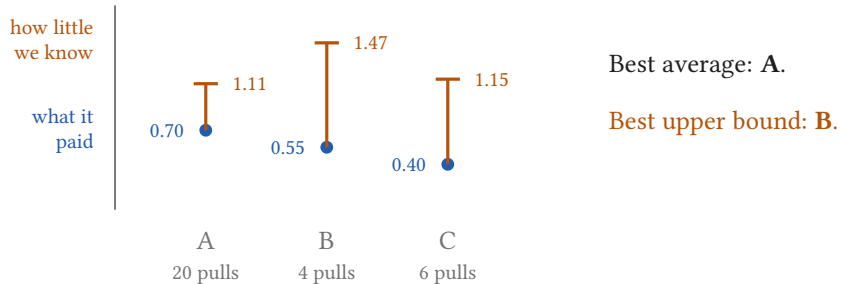
Bandits, and the upper bound



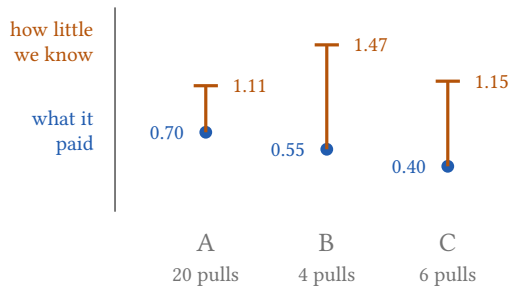
Bandits, and the upper bound



Bandits, and the upper bound



Bandits, and the upper bound

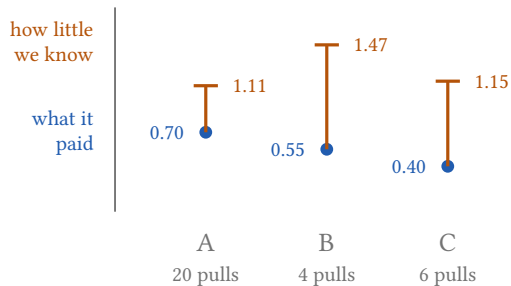


Best average: **A**.

Best upper bound: **B**.

Take **B**.

Bandits, and the upper bound



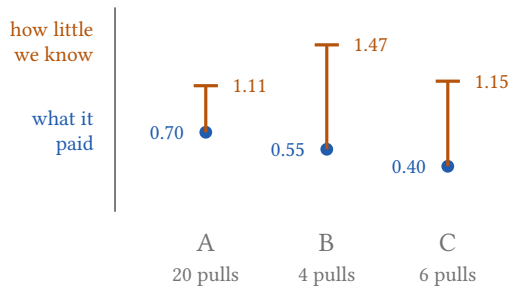
Best average: **A**.

Best upper bound: **B**.

Take **B**.

A tree picks a child the same way.

Bandits, and the upper bound



Best average: **A**.

Best upper bound: **B**.

Take **B**.

A tree picks a child the same way.

Play it



Section 2

The LATS method

2

UCT: the critic, plus an exploration bonus

$$\text{UCT}(s) = \underbrace{V(s)}_{\text{the critic}} + \underbrace{w \sqrt{\frac{\ln N(p)}{N(s)}}}_{\text{the exploration bonus}}$$

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The critic

what the model thinks
this branch is worth

UCT: the critic, plus an exploration bonus

$$\text{UCT}(s) = \underbrace{V(s)}_{\text{the critic}} + \underbrace{w \sqrt{\frac{\ln N(p)}{N(s)}}}_{\text{the exploration bonus}}$$

The critic
what the model thinks
this branch is worth

+

The exploration bonus
how little we have tried it

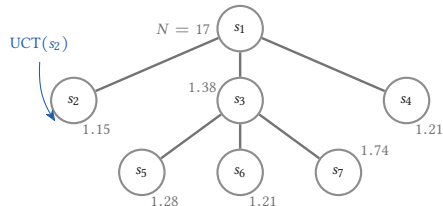
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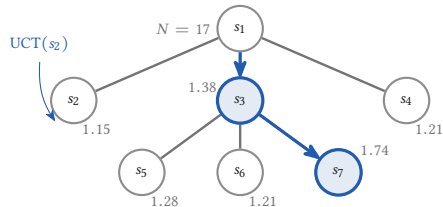


This morning's words for the same two terms: how good it looked, plus how little we know.

The same loop as this morning, with four edits



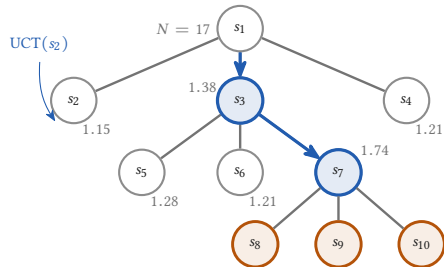
The same loop as this morning, with four edits



1 Selection

by UCT

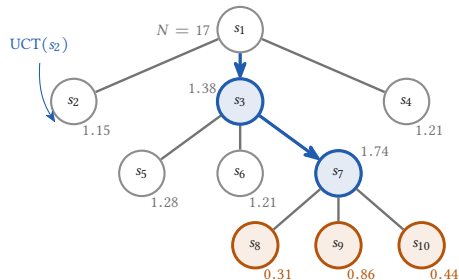
The same loop as this morning, with four edits



1 Selection by UCT

2 Expansion the *actor* proposes

The same loop as this morning, with four edits

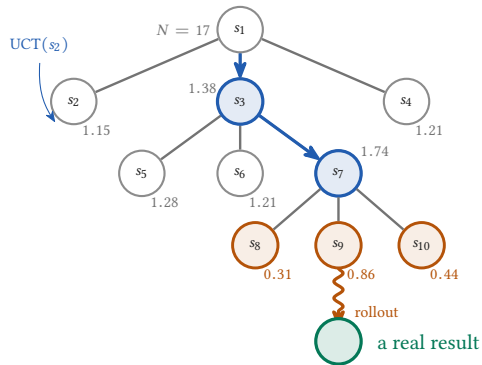


1 Selection by UCT

2 Expansion the *actor* proposes

3 Evaluation the *critic* scores

The same loop as this morning, with four edits



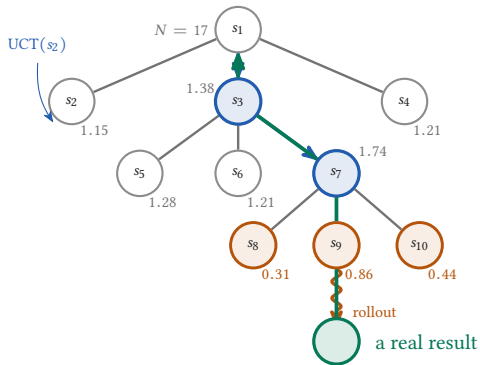
1 Selection by UCT

2 Expansion the *actor* proposes

3 Evaluation the *critic* scores

4 Simulation the *best* one, to the end

The same loop as this morning, with four edits



1 Selection by UCT

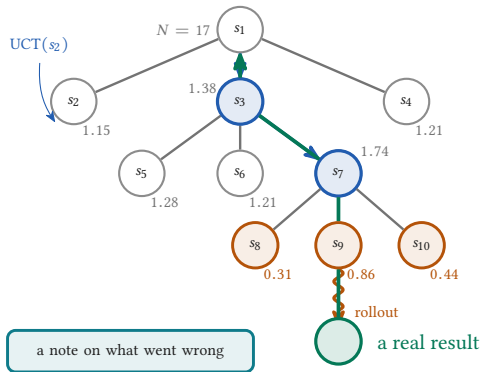
2 Expansion the *actor* proposes

3 Evaluation the *critic* scores

4 Simulation the *best* one, to the end

5 Backpropagation the *real* result, up

The same loop as this morning, with four edits



1 Selection by UCT

2 Expansion the *actor* proposes

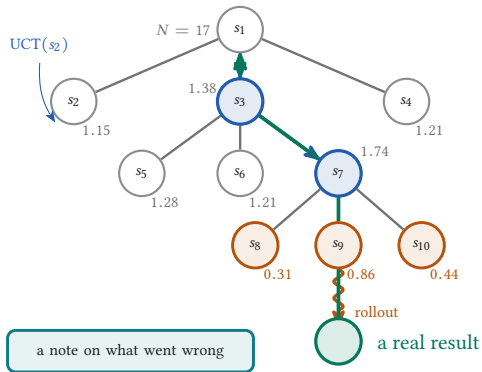
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6 Reflection the *feedback* model writes it

The same loop as this morning, with four edits



1 Selection by UCT

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3 Evaluation the *critic* scores

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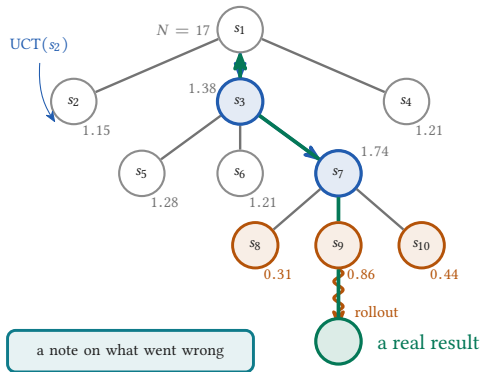
5 Backpropagation the *real* result, up

6 Reflection the *feedback* model writes it

grey: UCT, what selection compares.

amber: V from the critic, and all that is new.

The same loop as this morning, with four edits



1 Selection by UCT

2 Expansion the *actor* proposes

3 Evaluation the *critic* scores

4 Simulation the *best* one, to the end

5 Backpropagation the *real* result, up

6 Reflection the *feedback* model writes it

into the next prompt

grey: UCT, what selection compares.

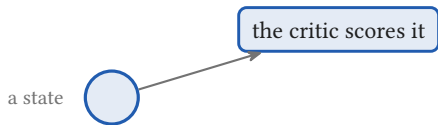
amber: V from the critic, and all that is new.

What the critic actually returns



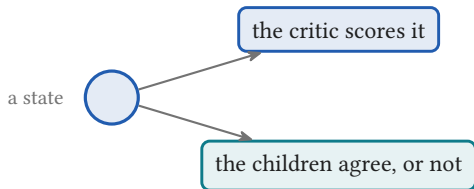
$$V(s) = \quad +$$

What the critic actually returns



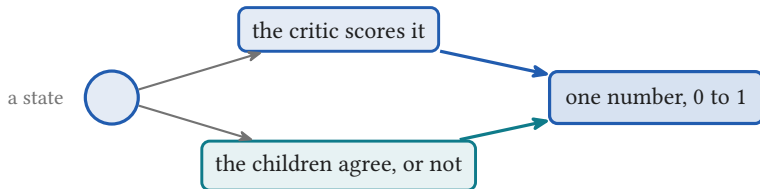
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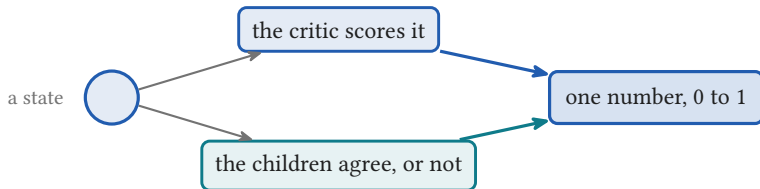
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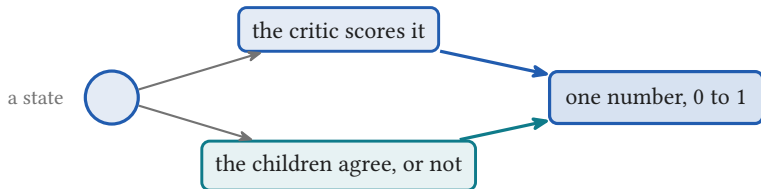
$$V(s) = \quad +$$

What the critic actually returns



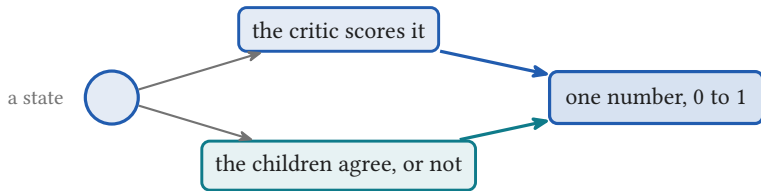
$$V(s) = \lambda LM(s) +$$

What the critic actually returns



$$V(s) = \lambda LM(s) + (1 - \lambda) SC(s)$$

What the critic actually returns



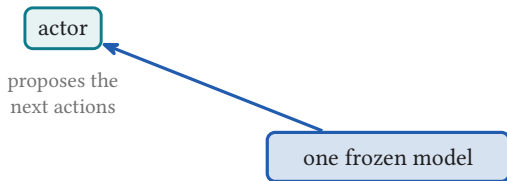
$$V(s) = \lambda LM(s) + (1 - \lambda) SC(s)$$

$\lambda = 0.5$ on question answering, 0.8 on programming and web tasks.

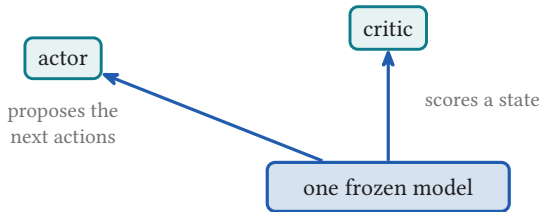
Three jobs, one set of weights

one frozen model

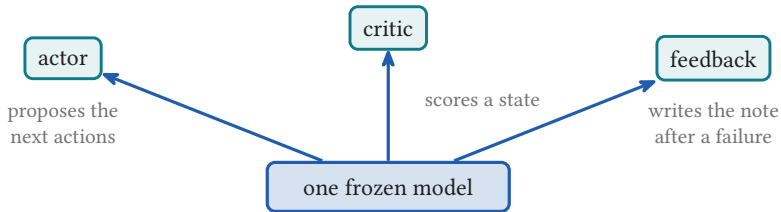
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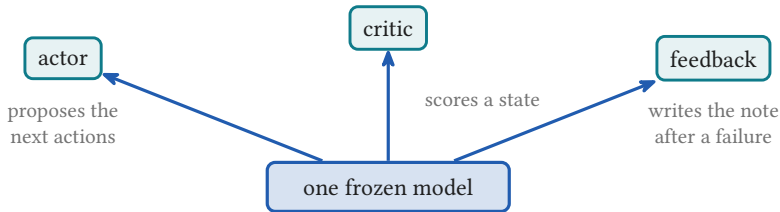
Three jobs, one set of weights



Three jobs, one set of weights



Three jobs, one set of weights



same weights every time. Only the prompt changes.